BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Year 10 Review and Year 11 Bridge

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Name the IDE facility used to write and change source code.

2. Which description matches an AND gate symbol?
   A) Triangle with output circle
   B) Curved input; pointed output
   C) Flat input; rounded output
   D) A decision diamond

3. "Adding 5 to a ticket count always calculates £5 per ticket." Correct this. Explain why.

4. (a) Convert denary 182 to two-digit hexadecimal. (b) Convert hexadecimal B6 to denary. Show your calculation.

Today's objectives

Identify specific gaps using assessment feedback. Improve answers using OCR command words and mark schemes.

Key vocabulary - Read, cover, recall

Command word
Instruction in an exam question, such as state, describe or explain.

Mark scheme
Guidance showing the knowledge and reasoning that earn marks.

Retrieval practice
Recalling knowledge from memory without looking at notes.

Trace table
Table recording variables and other values as an algorithm runs.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Know the two papers

Paper 1 is Computer Systems (sections 1.1-1.6); Paper 2 covers algorithms, programming, robustness, Boolean logic and languages (2.1-2.5). Each paper counts for half the GCSE.

Diagnose the missed mark

Look at the command word and the actual mark scheme. Ask whether the problem was missing knowledge, an incomplete explanation, a tracing error or a calculation mistake.

Improve the response

Write a technical point, then link it to the effect in the situation. 'RAM is fast' is vague; 'RAM holds running instructions and working data so the CPU can access them quickly' is a stronger explanation.

Make a retrieval plan

Choose two named specification points. Test without notes, compare with the mark scheme, correct mistakes, then answer a different question a few days later.

I Do - Worked model

A learner answers 'RAM is fast' to a question asking why RAM is used for currently running programs. Improve this into a full TWO-mark explanation. [2 marks]

- 'Explain' requires a point and a relevant reason, not only a vague adjective.
- RAM stores the instructions and data currently being used by the computer [1].
- The CPU can access that working information quickly when executing the running program [1].
- Both parts refer to the role of RAM. 'RAM is fast' alone would not secure both marks.

We Do - Guided practice

1. A learner needs a durable portable drive. Explain ONE reason an SSD may be preferable to an HDD. [2 marks]

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2. Trace this algorithm: total = 2; total = total + 3; print(total). State the total after both assignments and the output. [3 marks]

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3. A message claims to be from a bank and demands an immediate password reset through an unfamiliar link. Identify the likely attack and explain one safe response. [3 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the assignment that totals the three drive capacities.

```
array drivesGB = [64,128,256]
total = 0
for i = 0 to 2
...
next i
print(total)
```

You Do - Independent exam questions

Q2. A pupil has written a program that stores numbers in an array. Explain why a trace table can help identify a logic error. [3 marks]

Q3. A class uses a Wi-Fi network. Describe TWO actions the school can take to improve network security. [4 marks]

Q4. A school gives a student an assessment with 'state' and 'explain' questions. Describe how the student should respond differently to these two command words. [3 marks]

Exit Ticket - True or False?

Writing the name of a feature is always enough to gain two marks for an 'explain' question.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The editor.

2. C - Flat input; rounded output

3. Use total = tickets * 5; addition does not calculate £5 for each ticket.

4. (a) B6. (b) 11 x 16 + 6 = 182.

I Do - Explain the model

'Explain' requires a point and a relevant reason, not only a vague adjective.

RAM stores the instructions and data currently being used by the computer [1].

The CPU can access that working information quickly when executing the running program [1].

Both parts refer to the role of RAM. 'RAM is fast' alone would not secure both marks.

We Do - Model answers

1. An SSD has no moving parts [1], so it is less likely to be damaged by knocks when carried [1].

2. After first line total=2 [1]; after second line total=5 [1]; the output is 5 [1].

3. Phishing [1]; do not use the link [1] and contact the bank using a known official route to check the request [1].

Code First - Answer

```
array drivesGB = [64,128,256]
total = 0
for i = 0 to 2
total = total + drivesGB[i]
next i
print(total)
```

Expected output: 448

You Do - Self-mark

Q2: It records relevant variable or array values [1] as the algorithm executes step by step [1], so the point where a value first becomes incorrect can be found [1].

Q3: Use strong passwords or authentication [1] to reduce unauthorised access [1]. Encrypt the wireless connection [1] to protect data against interception [1].

Q4: 'State' generally requires a concise fact or answer [1]. 'Explain' needs a technical point [1] linked to a reason or effect [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.